

PRADEEP KUMAR BALASUBRAMANIAN

EMBEDDED SOFTWARE ENGINEER

Pune, India | Open to Relocation

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SUMMARY

Embedded Software Engineer with 4+ years of experience in C/C++, Python, and embedded firmware development. Experienced in automotive embedded systems, RTOS, communication protocols, bootloader development, and academic robotics projects.

WORK EXPERIENCE

Intangles Lab Pvt Ltd

Pune, India

Embedded Systems Engineer

Aug 2024 – Present

- Engineered telematics firmware and custom bootloaders on Renesas RA Series MCUs, implementing flash memory management for backup firmware images, update flags, and persistent configuration data.
- Architected Quectel EC200 (QuecOpen/QL_RTOS) firmware to manage end-to-end CAN polling, ECU communication, cloud routing, and FOTA workflows.
- Integrated and validated automotive communication protocols including CAN, J1939, ISO 15765-4 (UDS), XCP, K-Line, and J1708 for vehicle diagnostics.
- Performed system-level debugging, performance optimisation, and fault analysis to improve firmware stability and communication reliability.
- Collaborated with hardware, backend, and QA teams to support firmware validation, device scalability, and production readiness.
- Developed a Python application to automate ECU FOTA workflows.

Nanotronics Scientific Solutions

Coimbatore, India

Embedded Engineer

Jul 2020 – Aug 2022

- Developed production ready embedded firmware for IoT devices, industrial fuel dispensers, and home automation systems.
- Implemented and optimised multi-protocol communication using UART, MODBUS, SPI, and MQTT to ensure reliable real-time data exchange.
- Integrated FreeRTOS, designing tasks, queues, and timers for deterministic scheduling and efficient system performance.
- Conducted hardware-software debugging and system optimisation, improving product reliability and reducing field issues.
- Supported product bring-up, testing, and deployment across ESP32, ESP8266, STM32, and PIC platforms.

SAEINDIA

Bengaluru, India

Project Intern

Jul 2019 – Aug 2019

- Led electronics and embedded systems development for a Range-Extended Electric Vehicle (REEV) student project conducted in collaboration with General Motors.
- Contributed to embedded control logic, system integration, and vehicle performance validation.
- Collaborated with mechanical and powertrain teams for end-to-end system integration.
- Awarded Best Performance and Runner-up at a national-level competition.

EDUCATION

University of York

York, United Kingdom

Postgraduate Certificate, Intelligent Robotics

Sep 2022 – Dec 2023

- Key Projects: Autonomous Robotic Car, Robotic Arm, Rescue Robot (ROS).

Sri Shakthi Institute of Engineering and Technology

Coimbatore, India

B.E, Electronics and Communication Engineering

Jul 2016 – Nov 2020

- Key Projects: Range Extended Electric Vehicle.

SKILLS

Programming: Embedded C, C++, Python.

Embedded Platforms: Embedded Platforms: ARM Cortex-M (STM32, Renesas RA Series), Renesas RH850, ESP32, ESP8266, PIC, Arduino, Raspberry Pi, Quectel EC200 LTE Module.

RTOS & Embedded Systems: FreeRTOS, QL_RTOS (QuecOpen), Bootloader Development, Interrupt Handling, Low-Power Design, Firmware Reprogramming, Real-Time Systems.

Protocols: CAN, J1939, UDS (ISO 15765-4), XCP, K-Line, J1708, UART, SPI, I2C, MODBUS, MQTT.

Firmware & Debug: System Debugging, Hardware-Software Integration, Firmware Bring-up, Flash Memory Management, Bootloader Development, Performance Optimisation, Fault Analysis.

Tools & IDEs: Renesas e² studio, STM32CubeIDE, Keil µVision, Arduino IDE, Visual Studio Code, Git, GitHub, Logic Analyzer, Oscilloscope, MATLAB, ROS, AutoCAD.

Collaboration & Tracking: Jira, Confluence.

Operating Systems: Windows, Linux.

LANGUAGES

ENGLISH - Professional working proficiency

TAMIL - Native

HINDI - Professional working proficiency

JAPANESE - Beginner

INTERESTS

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- Embedded Systems Research, Automotive Telematics, Firmware Optimisation, Open-Source Contributions, Learning New Microcontroller Architectures and Robotics Embedded Development.

PUBLICATION

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- **Automotive Electronic System Controlling the Range Extended Electric Vehicle,** INTERNATIONAL JOURNAL OF ENGINEERING RESEARCH & TECHNOLOGY (IJERT) ICEECT – 2020 (Volume 8 – Issue 17)